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A Simple Tool to Assist in Agile Methodology Adoption Decisions

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ABSTRACT. *This paper discusses the importance of adoption factors to the adoption of an Agile method and the usefulness of a decision support tool to help determine the viability of such methods for specific software projects. It proposes the Adoption Assessment Matrix, to be used as a precursor to the selection and use of an Agile method. The Adoption Assessment Matrix was used to assess the suitability of Agile methods in software development projects in a series of workshops. The experiments show that the tool is beneficial as a guide for discussions, concentrating the debate on the critical factors, applied to the individual project, which proved as valuable to the decision making process as the output of the tool itself. Furthermore, the use of the Adoption Assessment Matrix moves beyond the debate on Agile versus traditional methodologies and concentrates on adoption decisions for individual projects, rather than on Agile as universal solution. It is often argued in the diffusion of innovation literature that the adoption of innovations can be hindered by the learning required to successfully deploy innovations. The decision support tool we propose attempts to overcome knowledge barriers by raising stakeholder awareness as to the suitability of Agile.*

RESUMÉ. *Cet article discute l'importance des facteurs d'adoption pour les méthodes Agiles et propose un outil pour aider les décisions d'adoption dans des projets informatiques précis. Nous proposons une matrice d'évaluation qui peut être utilisée au début du processus d'adoption. Cette matrice a été testée dans 2 études de cas d'adoption en entreprise. Nos expériences montrent que notre outil est bénéfique non seulement à cause des résultats qu'il génère, mais parce qu'il sert de guide pour les discussions entre décideurs. Les discussions peuvent ainsi évoluer d'un débat général pour ou contre Agile vers un débat ciblé sur le contexte du projet. Au total, notre outil d'aide se propose d'aider les participants à vaincre l'obstacle du manque de familiarité avec la méthode innovante en améliorant le dialogue sur le potentiel d'Agile pour un projet précis.*

KEYWORDS: *Agile Methods, Adoption Decision, Adoption Assessment Matrix, Decision Making.*

MOTS-CLÉS: *méthodes Agiles, décisions d'adoption, matrice d'évaluation, prise de décision.*

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1. Introduction

Discussions on Agile software development methodologies have a tendency to develop into an argument between proponents of Agile methods and proponents of more traditional process oriented methodologies. Indeed, proponents of both sides of the methodology debate give the appearance of being involved in a religious war. The title of an executive report from the Cutter Consortium sums this up: “CMM versus Agile Development: Religious Wars and Software Development” (Orr, 2002). Adding to this, Boehm and Turner (2004) describe the advocates of both approaches as *near-messianic*. The terminology used in these debates is unhelpful, and in many cases are inaccurate representations. For example, in order to promote an Agile concept, an attack on traditional hard methodologies is used; however, the guilt lies on both sides of the debate where advocates of the traditional/hard methodologies refer to exponents of Agile methods as hackers.

Problems seen with the introduction of a new software process, or methodology, can often be explained by the diffusion of innovation theory, described in Rogers (1962), and the variations of this theory by authors such as Fichman and Kemerer (1999), Green *et al.* (2004), Toleman *et al.* (2004), Gallivan (2001, 2004), and Agarwal and Prasad (2000). Although diffusion of innovations often refers to the diffusion of technologies, researchers have used the theory (and variations of the theory) to describe the introduction of software processes, methodologies, languages, etc. In fact, Swanson and Ramiller (2004, p. 554) borrowing Weick’s concept of *mindfulness*, suggested that “adopting organisations entertain scant reasoning for their moves. Especially where the innovation achieves a high public profile ... deliberative behaviour can be swamped by an acute urgency to join the stampeding herd, notwithstanding the high cost and apparent risk involved”. Indeed, Swanson and Ramiller (2004) argue that this *mindless* behaviour in pursuit of innovation is the rule, with the exception being a *mindful* organisation engaging in innovations. Bearing this in mind, the decision to adopt a particular software development methodology is a difficult one, and the decision to choose an Agile method is no exception.

When innovations (including process innovations) are diffused in an organisation, there are four phases (Swanson, 2001). Initially, there is comprehension of the innovation which is followed by adoption and then implementation. Finally, there is assimilation of the innovation. In the context of this paper it is the comprehension phase of such diffusion of innovation that is of most importance with regard to Agile method adoption decisions. However, there are no procedures available to assist decision makers choose the best development method for a given situation (Abrahamsson *et al.*, 2002). In the literature, both academic and practitioner, definitions and descriptions of various Agile methods are presented, but the factors that should be considered in the decision to adopt, or not adopt, an Agile method are not adequately addressed. This is not an uncommon trend in other areas of IS, for example: ERP adoption decisions, or the decision to adopt an e-business

model (Sammon and Adam, 2004; Hayes and Finnegan, 2003), and heightens the need for studying the importance of the intelligence phase of decision making.

Agile methodologies try to avoid the excessive use of tools (Beck and Fowler, 2001). However, one Agile methodology, Dynamic Systems Development Method (DSDM) does recommend the use of appropriate tools during the development process (Stapleton, 2003; Jones, 1998). Coesmans (2003, p. 119) agrees stating that “working from the mindset of DSDM (the principles) and using the right tools from the toolbox, gets many difficult projects successfully done”. In each case, although the tools are used to effect during the project, none are used effectively to assess the viability of the methodology, before project initiation. A review of the literature has highlighted that there are a small number of decision tools available to assist in this task of deciding whether to adopt an Agile method or not (Boehm and Turner, 2003; 2004; Pikkarainen and Passoja, 2005). Boehm and Turner (2004) provide a graphical method to assess various aspects of organisational suitability for Agile and Pikkarainen and Passoja (2005) provide a case study example of how to conduct an assessment to determine which of the Agile practices is suitable for a product development organisation. However, both of these assessment methods fall short in increasing our understanding of the successful adoption of Agile methods, especially by those considering adoption. This limitation in use is due to the fact that a comprehensive review of all factors critical to the adoption of an Agile method is not provided (for example, Boehm list only 5 critical Agile/disciplined decision factors), furthermore, the benefit of generating consensus from creative conflict in a workshop environment is not a feature of these approaches to assessment. Finally, while Boehm and Turner (2004) provide a method to determine the current state of an organisation’s agility and Pikkarainen and Passoja (2005) provide an insight into the processes to use when adopting an Agile approach, neither of these assessment methods focus on gathering the perceptions of managers and developers (captured in the form of a confidence rating) to assess if an Agile approach would be viable in the context of their software projects.

The next sections describe what we mean by Agile methods and we provide a number of critical adoption factors that can be considered when deciding whether to adopt an Agile methodology. These critical adoption factors are the key component of an Adoption Assessment Matrix that can be used by software teams to determine the viability of adopting Agile methods. We then describe two workshops where the Adoption Assessment Matrix was used by software developers in a commercial setting, and one workshop held with development staff from the British Ministry of Defence. Our results are presented with conclusions drawn from the workshops. In the final section we present recommendations for future research.

2. Agile software development methodology

To describe “the Agile method” is a misnomer. The Agile software development method does not exist; it is instead a collection of methodologies with common core values, where examples of Agile approaches include: Extreme Programming (XP); Crystal Methods; SCRUM¹; Dynamic Systems Development Method (DSDM); Feature Driven Development (FDD); and Adaptive Software Development (ASD) (Highsmith, 2001; Sutherland, 2001). For many proponents of the Agile methodologies, the epoch of the Agile movement was February 11th 2001, when representatives of the different Agile methodologies convened in the mountains of Utah to create the Manifesto for Agile software development. The Agile manifesto, available at <http://Agilemanifesto.org>, is a collection of values that underlie all Agile methodologies:

- Individuals are more important than processes and tools;
- Working software is more important than comprehensive documentation;
- Customer collaboration is more important than contract negotiation;
- Responding to change is more important than following a plan.

Therefore, Agile methods are a response to the inability of traditional methods to embrace change in a turbulent business environment, that demands software to meet its needs quickly (Highsmith and Cockburn, 2001; Kruchten, 2001), and Rising and Janoff (2000, p. 3) describe it as the need to “meet customer needs and turn this chaos to our advantage”. The manifesto, its origins, and its importance to the Agile methods are discussed in a variety of research including: Lindvall *et al.* (2002), Fowler and Highsmith (2001), Boehm and Turner (2004), Highsmith (2004), and Koch (2004).

Glass (2001) describes the debate between proponents of traditional development approaches and proponents of the newer Agile approaches, arguing that both sides of the debate have merit and demerits. Neither approach is correct in all circumstances, and the “best-fit” needs to be determined for a given circumstance. According to Brookes (1986) there are no “silver bullets” in software development, and there probably never will be. Of all of the Agile methods, DSDM, (1998) comes closest to recommending an approach for determining the viability of an Agile development method for a project, albeit with a limited look at a feasibility study. DSDM recommends that the organisation has the right culture for using Agile approaches, but it is not specific about how one should go about measuring or evaluating this culture. Furthermore, the steps in the feasibility study that follow this recommendation involve educating a key stakeholder, and producing a strategy and plan. Based on these recommendations, it appears that the choice has already been made to adopt DSDM, so it is not a decision whether to adopt an Agile approach or not.

1. SCRUM is not an abbreviation, it is a rugby term used to depict a strategy meeting which gets an out of play ball, back into play.

Williams *et al.* (2000) state that there are a variety of software projects and a variety of software processes to match the projects' goals and objectives. However, on examination, there is no discussion amongst advocates on what defines a project as being suitable for this Agile technique, as opposed to any of the plethora of other software development techniques, processes, or methodologies. Therefore, the debate concerning Agile methodologies has predominantly been based around whether it was a better choice than traditional development methods (De Marco and Boehm, 2002), rather than a debate on the appropriateness of an Agile approach for a given company, team, or project.

2.1. Critical adoption factors in the decision to adopt an agile methodology

Throughout the available literature on Agile methodology, factors important to the success of an Agile project are discussed, yet these factors are not specifically used to determine the viability of adopting an Agile approach. For example, the Agile manifesto, described in Abrahamsson *et al.*, (2002), is a list of aspirations or ideals, and as such is not readily quantifiable as a method of adoption assessment. Therefore, as a result, having conducted a preliminary literature review for the purpose of this research, and based on the industry experience of one of the researchers involved (in the adoption of Agile methods in software projects), we present ten factors, and their descriptions, that could be regarded as critical adoption factors which can form the basis for a discussion when assessing the suitability of a software project to the adoption of an Agile methodology, and indeed the agility of an organisation, as illustrated in Table 1.

The researchers acknowledge that there may be differences of opinion as to what should constitute the ten critical adoption factors, but, are also aware that these factors could change through further research. Like any list, for example, be it top ten athletes or favourite music, there will never be universal agreement on its constituent parts. The list is not set in stone, but an important starting point. In the context of the critical adoption factors highlighted in Table 1 we borrowed the concepts of Fichman and Kemerer (1999) where they list the structural, managerial, political and social factors that impact the diffusion of innovation and we further categorised the ten critical adoption factors into four decision making areas, namely: Project, Customer, Team, and Organisation. These four groups represent the critical elements of any software project in terms of its suitability to the adoption of Agile methods, namely, the project itself, the team assigned to the project, the client for whom the project is being delivered, and the organisational setting in which the project is undertaken.

Table 1. *Critical Adoption Factors for an Agile methodology*

Critical Adoption Factors	Group	Description	References
Duration of the project	Project	The timeframe for the project should be short for an Agile project	Highsmith (2001) Kruchten (2001)
Location of the customer	Customer	It is preferable that customers are available throughout the project, therefore an in-house customer would be ideal	Young (2003) Jeffries (2001)
Customer Involvement	Customer	Customer involvement is vital for the success of a project	Beck and Fowler (2001) Young (2003)
Acceptance of change (to requirements)	Project	Agile methods are specifically aimed at projects subject to continual change	Highsmith (2001)
Team Size	Team	Agile methodologies emphasise the importance of teams, recommending smaller team sizes	Boehm (2002) Rising and Janoff (2000)
Skill level of team	Team	Highly skilled developers are required	Reifer (2002) Levine <i>et al.</i> (2002) Lindvall <i>et al.</i> (2002)
Organisational and reporting structure	Organisation	An organic structure is required for an Agile approach	Boehm (2002) Cockburn and Highsmith (2001) Dolan <i>et al.</i> (2003)
Process	Organisation	Processes, such as CMM or TL900, are regarded by many as stifling, even stopping, Agile methodologies. At the very least, they can be in conflict	Martin (2003) Greening (2001)
Documentation requirements	Organisation	While Agile methodologies do not prohibit documentation, it should be kept to a minimum	Greening (2001) Olson and Stimmel (2002)
Layout of workspace	Organisation	Open planned offices, with shared areas, are required to promote communication and team work	Poole and Huisman (2001); Kalita (2003)

2.1.1. *Project*

The project grouping describes the relevance of critical adoption factors when considering the mechanics of the project being undertaken. Issues of importance that can impact on the appropriateness of adopting Agile methods include: the duration of the project; increasing uncertainty and changing requirements within the project. “Today’s project manager must deliver concrete results in shorter time frames while

being constantly bombarded with myriad changes and risk laden decisions” (Highsmith, 2001, p. 260). This concern aligned with Kruchten’s (2001) criteria of reacting quickly to market, impose a shorter time frame on Agile projects. A core value of Agile development is the acceptance of change, usually visible through changing requirements. Highsmith (2002) proposes that as the uncertainty associated with a project increases, the suitability of the Agile approach will also increase. Kirkpatrick *et al.* (1992) describe the changing of requirements (requirements volatility) as one of the major sources of risk in a project. However, this directly contradicts the Agile viewpoint, where stories, requirements definitions in the Agile methodology, continue to be written throughout the project. This is regarded as one of the basic edicts of the Agile approach (Beck and Fowler, 2001). Burns and Stalker (1961) found that firms could be categorised into two distinct management types, which were the best fit for their working environment. Mechanised systems are appropriate for stable environments, while organic systems are pertinent to conditions of change and flux. Although this categorisation was proposed years before the advent of Agile methods, projects using Agile methodologies show a close correlation with the definition of organic systems.

2.1.2. *Team*

The team grouping highlights the importance of the project team, in terms of team size and skill level of the team, for adopting Agile methods. While the team size is important, it is not of vital importance. Boehm (2002) describes how Agile development is optimal when used in small teams. Rising and Janoff (2000) quantify ten as the optimum number of developers, although not always compulsory, in an Agile team. The skill level of the team is a continually stated requirement in research into Agile methods. Martin (2003) further defines skill level by stating that a strong player does not necessarily have to be an expert programmer, though they must work well with others. Good communication skills and an ability to interact with others are of higher importance than expertise in a programming language. Furthermore, Reifer (2002) makes an interesting observation, which may have repercussions for Agile research and implementation. Reifer (2002) surveyed thirty-one Agile projects across eight industry sectors and comments that the teams involved were made up of motivated, experienced programmers, in cohesive teams. Furthermore, Levine *et al.* (2002) believe that Agile projects must be staffed with skilled, experienced, software developers, while Lindvall *et al.* (2002) make a similar observation by discussing the percentage of an Agile team that needs to be experienced or composed of “good” people. As these descriptions do not apply to every developer and team in the software industry, there may be issues with Agile if implemented in non-exceptional teams.

2.1.3. *Customer*

The customer grouping describes the customer, or customers, of the project, identifying location of the customer and customer involvement as critical adoption

factors to consider in adopting an Agile methodology. In fact, Turk *et al.*, (2005) describe customer involvement as one of the core assumptions for XP. Beck and Fowler (2001) are willing to compromise on the definition of customer. The customer is the person who makes the business decisions *i.e.* completion date, scope, etc (Beck and Fowler, 2001). This person can be an internal product manager or an individual who will purchase the software. Young (2003) describes how a DSDM project in British Airways addressed the necessity for user involvement in DSDM projects. The project was in the area of e-commerce, so its user base was extensive and difficult to categorise. British Airways used sales and marketing staff to represent the customers. Furthermore, Jeffries (2001) specifically use the term customer as opposed to customers, as there is a requirement for a single customer voice, which implies difficulties for projects with multiple customers in multiple regions. However, Reifer (2002) shows that the theory of customer participation is not reflected in real Agile projects, in that customer surveys are used as a replacement, and attempts are made to get customer representatives on site as often as possible. In addition, Beck and Fowler (2001), while accepting the various definitions of customer, or customer stand-in, declare that a customer must be an integral part of the team. Without this integration into the team, the project will fail.

2.1.4. Organisation

The organisation grouping highlights the criticality of the organisational environment, which comprises the following: organisational and reporting structure; process; documentation requirements; and layout of workspace. Boehm, (2002) states that Agile requires both responsive people and organisations. The relationship between managers and developers is one of collaboration rather than the traditional command and control structure (Cockburn and Highsmith, 2001). “The inefficiency of rigid bureaucratic structures, with many hierarchical levels and watertight compartments, can no longer be tolerated in companies that must compete in turbulent environments” (Dolan *et al.*, 2003, p. 28). Also, Reichheld (2001), although discussing loyalty in the workforce, makes the pertinent point that in the complex work environment that exists, organisations need to be adaptable. To ensure this adaptability, managers must simplify the rules of decision making, adopting an organic system of management, as defined by Burns and Stalker (1961).

It is clear that process, or the lack of process, plays a significant role in Agile methods (Highsmith, 2001). Discussing the Agile concept of the importance of individuals over process highlights that “a good process will not save the project from failure if the team doesn’t have strong players” (Martin, 2003, p. 4). Therefore, the absence of rigid processes is embraced. According to Gandossy (2003, p. 32) “fast companies have a defining sense of purpose, supported by a few simple rules. They determine what is essential and ignore the rest”. For example, Greening (2001) describes the difficulties involved in the implementation of XP, in a company that has a formal software development process.

A variation, or continuation, of the discussion on process is the documentation requirements of an Agile project. Greening (2001) stresses that XP does not prohibit documentation, it merely stresses that documentation has a cost. Any documentation to be used must be evaluated to ensure that it has a benefit and that it is definitely necessary. This is based on the assumption that tacit knowledge is more valuable than explicit knowledge (Turk *et al.*, 2005). The usefulness of documentation, as a method of controlling a project, is disputed by Johansen and Perkins (2002). The argument against XP was that documentation is required for the future engineers who will maintain the software product. “Without good documentation, you know you will suffer undue loss if you lose staff” (Olson and Stimmel, 2002, p. 11). The layout of the workplace receives considerable attention in literature on Agile methods. Poole and Huisman (2001) describe the efforts made in Iona Technologies to enhance team communications. The previous environment they describe resembles that of the Dilbert cartoon. The multiple bays of shoulder height cubicles were replaced by a common area, or group workspace. This was an attempt to increase awareness of the team, as opposed to individuals in their own tiny spaces. Kalita (2003) demonstrates that co-located teams, although not a formal principal of DSDM, should be used in DSDM projects to facilitate the necessity of communications.

3. Data collection and results

The critical adoption factors are used to provide a simple structure to assist companies in the Agile method adoption decision making process, through the use of an “Adoption Assessment Matrix” (Table 2). These factors are not ordered or grouped, as it is not intended to lead the individual completing the matrix. However, each factor does relate back to the values of the Agile manifesto, for example, the process factor relates to the importance of individuals over processes. Furthermore, one overriding assumption made throughout this research is that “life-critical projects” are not suited to an Agile approach, an observation also made by Boehm and Turner (2003, 2004).

The Adoption Assessment Matrix was used in Agile methods workshops organised by the researchers in two software development companies (Company A Telecommunications and Company B Engineering) and with personnel from the British Army, Navy and Air Force (tri-service). The two companies were selected because they operate in different industrial sectors: Company A Telecommunications and Company B Engineering. The tri-service workshop was held to see whether developers from very process oriented organisations would be confident to adopt Agile methods. It was hoped that the differences between the groups selected to participate in the workshops would provide a diverse view of the possibilities in adopting an Agile methodology in practice.

Table 2. *Adoption assessment matrix*

Critical adoption factor	Weighting	Rank Case	Result Case
Duration of the project 1 = more than 5 years 5 = less than 6 months	4		
Location of the customer 1 = many customers in many countries 5 = in house	4		
Customer involvement 1 = will have no interaction 5 = willing to interact	4		
Acceptance of change (to requirements) 1 = rigid 5 = flexible	4		
Team size 1 = more than 20 5 = up to 3	3		
Skill of team 1 = inexperienced 5 = very experienced	3		
Organisational and reporting structure 1 = many reporting layers 5 = flat structure	2		
Process 1 = 5 or more standards to follow 5 = no standards to follow	2		
Documentation requirements 1 = a lot of documentation required 5 = very little documentation required	1		
Layout of Workspace 1 = individual cubes, people isolated 5 = open plan, no walls	1		
Confidence Rating	28		
Note on Workings of Matrix: [1] Rankings (between 1 and 5) are applied to each of the critical adoption factors. These rankings are based on a workshop participants assessment of the factors applicability to a particular project. [2] Each ranking is multiplied by the weighting for each critical adoption factor, providing a result for each critical adoption factor. [3] Finally, the totalled output, referred to as the “confidence rating” is calculated as follows: Confidence Rating Calculation = (SUM(Result Case)/(28*5))*100			

In the workshops for the two commercial software development companies the participants were using the Adoption Assessment Matrix as an actual decision support tool, in that they were using their practical development experiences to assess the viability of adopting an Agile method for their specific software development projects. It is worth noting that a major benefit of the tool is that it guides discussion, concentrating the debate on the critical factors, once applied to the individual project. These discussions proved to be as valuable in the decision making process as the output of the tool itself. Developers and managers completed the Adoption Assessment Matrix independently, so as to provide a broader perspective from each company of those comprising the project team, and to present possibilities to the researchers to further refine the individual critical adoption factors, and of the groupings of factors. It is important to note that the derived confidence rating is not compared to a threshold, used to recommend or reject the adoption of an Agile approach. However, the confidence rating is used to facilitate debate, which in turn leads to a better analysis of suitability; for example, the matrix as completed by managers and developers of a project team highlights differences in the confidence ratings, which encourages debate.

In total thirteen adoption assessment matrices were completed by workshop participants from the two software development companies, and eight adoption assessment matrices were completed by the tri-service personnel. These were analysed in order to assess the participants' perception of the viability of the adoption of an Agile method, with the view of also improving the Adoption Assessment Matrix for future use. A summary of the completed matrices for Company A and Company B and the tri-service case is presented in Table 3.

Table 3. *Summary of the three workshops*

	Company A		Company B		Tri-service
Critical adoption factors	Average developer	Average manager	Average developer	Average manager	Average
Confidence rating	55	60	73	81	66

The researchers used the tri-services case as an instrumental case (Stake, 2000) to provide insight into certain issues around the decision to adopt an Agile method. While the workshop format was identical to that conducted in the two commercial software development companies, the case was of secondary interest, playing a supportive role; the purpose of which was to facilitate our understanding of issues around: the criticality of a project, the documentation required for a project, the processes to follow in completing a project, and the timeframe for completion of a project; and the impact of these issues on the decision to adopt an Agile method. The researchers' initial expectation of the tri-services case was one of unsuitability to

adopt an Agile method, due to the highly process-oriented nature of the British Ministry of Defence.

Discussion

The most striking observation, from the analysis of the two commercial software development companies was that managers were more positive about the benefit of adopting an Agile methodology than the developers themselves. The average confidence ratings for managers against developers were 60% compared to 55% (for Company A) and 81% compared to 73% (for Company B). This seems unusual as we expected developers to appreciate the Agile methodologies more than managers, due to the fact that Agile methodologies propose a movement away from rigid management to one of increasing trust in and empowerment of developers. For example, definitions of management as described in Koontz *et al.* (1980); Lucas (1982); and Cole (1988) would imply an approach that, while not anti-Agile, could be considered as presenting problems for the use of Agile methods. In fact, management often resist the introduction of an Agile method as they have a variety of concerns, generally concerning the lack of process (Rasmusson, 2003). Karlstrom and Runeson (2003) describe how management in one organisation regarded the Agile methodology used as a threat and disbanded the project. Therefore, it is interesting to find managers being more optimistic and positive about an Agile approach than developers (who some consider Agile to be aimed at). One possible reason is that Agile is regarded by developers as just another set of processes aimed at forcing a team to comply with policy, as suggested by Birmingham (2002). The impact of this, from the viewpoint of the Adoption Assessment Matrix is that the role of the individual completing the matrix needs to be taken into consideration (e.g. accepting that managers may be overoptimistic and/or developers may be overly pessimistic). It is noted though, that the difference in confidence ratings between managers and developers is not that large.

Analysing the groupings of adoption factors provides additional feedback for use in the decision making process. The greatest difference between developers and managers arises from their views on the rankings applied to the organisational group of adoption factors. This was the greatest difference in both software development companies. The organisational factors concern the organisational environment in which projects take place. Managers are more positive, in both companies regarding this area. This may be explained by the fact that organisational factors encompass areas of control: process, documentation, etc. These areas are predominately management requirements, and are quite often resented by developers. It is in the area of control, and the level of control, that management and development often differ, and “the inescapable interface between managers and their employees is the control process” (Storey, 1983, p. 83). Again, this is an area that will have an impact of the Adoption Assessment Matrix. The level of control in a company will need to

be further examined within the structure of the matrix, to better determine why developers specifically give a lower weighting.

The area where developers were more positive than managers was that of the skill of the team. It is probably obvious that developers would have a higher opinion of their own ability than managers, but, from the point of view of the decision making process, the role of the individual completing the survey has an impact. The overall average confidence rating for both companies shows a large difference. The confidence rating is an indication of the suitability of an Agile methodology for this particular development team, for a particular project. Company A had an average confidence rating of 57.5% while Company B had a confidence rating of 77%. Knowledge of the two companies explains an aspect of this difference, but not completely. The main factor in this difference is with the organisational group of critical adoption factors. The matrices from Company A show that there is a higher degree of process required, and this was verified through discussions with the team. Company B is a considerably smaller organisation than Company A, so there is a lower level of process. The Adoption Assessment Matrix does highlight this fact, so the importance of the organisational factors in the matrix is shown. Furthermore, the difference in the customer grouping reflects the nature of the individual projects undertaken; where a project in Company A serves multiple customers, as opposed to a project serving a single customer in Company B. This explains the perception of customer involvement in a project, within Company A, which is identified in the lower rating attributed to the customer factors.

Overall, the analysis of the matrices completed by the two software development companies highlights the benefit of the use of the Adoption Assessment Matrix. Although, the output of the tri-services workshop was expected to reveal rather different results from the commercial cases due to the process-intensive nature of the organisation, some interesting, if not divergent, issues were discussed. The respondents overall average confidence rating was 66% which was higher than the confidence ratings returned by Company A, and somewhat lower than the ratings returned by Company B. Again, the usefulness of the actual discussions around the ratings of the Critical Adoption Factors facilitated the participants to consider stages of a project, as opposed to the entire lifecycle of the project. The participants explained that they were reasonably confident that Agile methods would work for the development stage of projects, but qualified this by stating that the initial feasibility, planning and analysis stages would favour process-oriented techniques. For example, projects within the Ministry of Defence go through a chain of command, however, it was also highlighted that some of these projects are often fast-tracked for completion in a shorter time-frame, and as a result, it was identified that Agile methods would be suitable in such a situation. The participants also confirmed our overriding assumption that life-critical projects are not suited to an Agile approach.

4. Conclusion

Agile methods have been described as glorified hacking by those who argue against their adoption and use. A more structured approach to the decision to adopt an Agile method would help to dispel these views. Some work has been carried out highlighting this argument; for example, Svensson and Host (2005) investigated the introduction of an Agile method and concluded that an assessment should have been done before the project. Although the Agile methodology recommends limited use of tools, the purpose of this Adoption Assessment Matrix is to support decision makers in assessing the suitability of adopting an Agile approach to software development. Therefore, we consider the use of this decision support tool to be a precursor to the selection and use of an Agile methodology. The use of the Adoption Assessment Matrix moves beyond the debate on Agile versus traditional methodologies and concentrates on specific cases of methodology adoption, rather than on Agile being the universal solution. The three cases involved in this research study using the Adoption Assessment Matrix believed that the matrix made them think differently about their decision, and provided a solid foundation for that decision. It is not proposed that the Adoption Assessment Matrix provides the ultimate answer in making that decision, but its use does provide immediate benefits. Universal agreement on the list of factors, and their weightings, is not possible, however what is important is that they are the basis for the debate on the suitability of an Agile method for a particular team. Further, the Adoption Assessment Matrix need not be a stand-alone tool. A variety of other tools can be utilised to assist in the adoption decision, for example a formalised approach, such as GQM – Goal Question Metrics (Basili *et al.*, 1994), or an approach to assess which blend of Agile approach to use (Pikkarainen and Passoja, 2005), can be used in addition to the Adoption Assessment Matrix presented here.

As managers look to Agile methods as a solution for the myriad of problems inherent in software development, there can be a rush to judgment, where an assumption is made that an Agile approach will work for a project. Projects are rarely assessed prior to the adoption of an Agile method. Ultimately this will increase the likelihood of problems and ultimately the failure of projects. However it is worth considering that problems that occur during Agile projects could have been predicted if a proper assessment had been conducted and steps taken to mitigate these potential problems. On the other hand, an assessment could determine that an Agile method would be unsuitable for a project, and considerable time and effort could be saved. The debate is moving on from pro- and anti-Agile to one of the acceptance of Agile in certain circumstances. The view of Agile as a ‘silver bullet’ should not exist so there is a necessity for decision makers to determine which projects could benefit from an Agile approach and which projects need a more traditional process based approach.

Finally, the use of the Adoption Assessment Matrix (decision support tool) in a workshop environment is extremely important to ensure that adequate planning and

the associated level of understanding of potential consequences exist to govern decisions around the adoption of an Agile method. The workshop is of practical relevance to organisational decision-makers and promotes the importance of the intelligence phase of decision making (reminiscent of Klein and Associates' pre-mortem sessions or the proper recognition of the problem in Klein's Recognition-Primed Decision (RCP) model) (Klein, 1993, 2002). Therefore, it is proposed that a workshop environment will promote the establishment of a more *mindful* (Swanson and Ramiller, 2004) approach to Agile method adoption through debate and creative conflict among workshop participants.

Recommendations for future research

Applying an "engineering" approach to the adoption decision of Agile methods in no way diminishes the core values of the Agile approach. Although the Agile methodology recommends limited use of tools with the project, the purpose of this Adoption Assessment Matrix is to support decision makers in (a) identifying the current state of an organisation's agility, and (b) assessing the suitability of adopting an Agile approach to a software development project and the likelihood of success. The factors and their weightings will evolve over time – this is to be expected as Agile itself is evolving with variations on existing methods appearing and disappearing. Therefore, the next phase of this research study involves the researchers taking the decision support tool (Adoption Assessment Matrix) into additional organisations to further refine its comprising critical adoption factors, their weightings, and the overall design of the decision support tool (the inclusion or exclusion of certain critical adoption factors). The researchers believe that with further refinement, the Adoption Assessment Matrix has the potential to be further successfully utilised in companies deciding to adopt an Agile methodology.

5. Bibliography

- Abrahamsson P., Salo O., Ronkainen J., Warsta J., *Agile software development methods, Review and analysis*, VTT Publications, Finland, 2002.
- Agarwal R., Prasad J., "A field study of the adoption of software process innovations by information systems professionals", *IEEE Transactions on Engineering Management*, Vol. 47, No. 3, 2000, p. 295-308.
- Basili V.R., Caldiera G., Rombach D., "The Goal Question Metric approach", Marciniak J. (ed), *Encyclopedia of Software Engineering*, Wiley & Sons Inc, 1994, p. 528-532.
- Beck K., Fowler M., *Planning extreme programming*, Addison Wesley, NJ, USA, 2001.
- Birmingham D., *Software development on a leash*, Springer-Verlag, NY, USA, 2002.
- Boehm B., "Get ready for Agile methods, with care", *IEEE Computer*, Vol. 35, No. 1, 2002, p. 64-69.

- Boehm B., Turner R., "Rebalancing your organizations agility and discipline", *Extreme programming and Agile methods – Xp/Agile universe*, Maurer F., Wells D. (eds), Springer-Verlag, Berlin, Germany, 2003, p. 1-8.
- Boehm B., Turner, R., *Balancing agility and discipline*, Pearson Education, MA, USA, 2004.
- Brookes F., "No silver bullet - essence and accident in software Engineering", *Proceedings of the 10th IFIP World Computing Conference*, 1986, p. 1069-1076.
- Burns T., Stalker G., *The Management of Innovation*, Tavistock Publications, London, England, 1961.
- Cockburn A., Highsmith J., "Agile software development: The people factor", *IEEE Computer*, Vol. 34, No. 11, 2001, p. 131-133.
- Coesmans P., "DSDM in a non-IT project", *DSDM Business focused development*, Stapleton J. (ed), Pearson Education. London, UK, 2003, p. 113-119.
- Cole G., *Management Theory and Practice*, DP Publications, Hants, England, 1988.
- De Marco T., Boehm B., "The Agile methods fray", *IEEE Computing*, Vol. 35, No. 6, 2002, p. 90-92.
- Dolan S., Garcia S., Auerbach A., "Understanding and managing chaos in organisations", *International Journal of Management*, Vol. 20, No. 1, 2003, p. 23-35.
- DSDM, *Guidelines for the introducing DSDM into an organisation*, DSDM Consortium, Kent, England, 1998.
- Fichman R. and Kemerer C., "The illusory diffusion of innovation: An examination of assimilation gaps", *Information Systems Research*, Vol. 10, No. 3, 1999, p. 255-275.
- Fowler M., Highsmith J., "The Agile manifesto", *Software Development*, Vol. 9, No. 8, 2001, p. 28-32.
- Gandossy R., "The need for speed", *Journal of Business Strategy*, Vol. 24, No. 1, 2003, p. 29-33.
- Gallivan M., "Organizational adoption and assimilation of complex technological innovations: Development and application of a new framework", *DATABASE*, Vol. 32, No. 3, 2001, p. 51-85.
- Gallivan M., "Examining IT professionals' adaptation to technological change: The influence of gender and personal attributes", *DATABASE*, Vol. 35, No. 3, 2004, p. 28-49.
- Glass R., "Agile versus traditional: Make love not war", *Cutter IT Journal*, Vol. 14, No. 12, 2001, p. 12-18.
- Green G., Collins R., Hevner A., "Perceived control and diffusion of software process innovations", *Journal of High Technology Management*, Vol. 15, No. 1, 2004, p. 123-144.
- Greening J., "Launching extreme programming at a process-intensive company", *IEEE Software*, Vol. 18, No. 6, 2001, p. 27-33.
- Hayes J., Finnegan P., "Assessing the potential of e-business models, Towards a framework for assisting decision-makers", *European Journal of Operational Research*, Vol. 160, No. 2, 2003, p. 365-379.
- Highsmith J., "Opening Statement", *Cutter IT Journal*, Vol. 14, No. 12, 2001, p. 2-4.

- Highsmith J., *Agile Software Development Ecosystems*, Addison Wesley, MA, USA, 2002.
- Highsmith J., *Agile Project Management*, Pearson Education, MA, USA, 2004.
- Highsmith J., Cockburn A., "Agile software development: The business of innovation", *IEEE Computer*, Vol. 34, No. 9, 2001, p. 120-122.
- Jeffries R., "Card magic for managers: low-tech techniques for design and decisions", *Beyond chaos: The expert edge in managing software development*, Constantine L. (ed), Addison-Wesley, NJ, USA, 2001, p. 27-32.
- Johansen K., Perkins A., "Establishing an Agile testing team", *Extreme Programming and Agile methods – XP/Agile Universe Chicago*, Springer-Verlag, NY, USA, 2002, p. 52-59.
- Jones C., "Software Project Management in the 21st Century", *American Programmer*, Vol. 11, No. 2, 1998, p. 24-29.
- Kalita T., "DSDM in process improvement", *DSDM Business focused development*, Stapleton J. (ed), Pearson Education, NY, USA, 2003, p. 175-191.
- Karlstrom D., Runeson P., "Scaling Extreme Programming in a market driven development context", *Proceedings of the 4th International Conference in Extreme Programming and Agile processes in software engineering (XP 2003)*, Genoa, Italy, 2003, p. 363-365.
- Kirkpatrick R., Walker J., Firth R., "Software development risk management: an SEI appraisal", *SEI Technical Review*, Software Engineering Institute, PA, USA, 1992.
- Klein G.A., "A recognition-primed decision (RPD) model of rapid decision making", *Decision Making in Action, Models and Methods*, Klein G.A., Orasanu J., Calderwood R., Zsombok C.E. (eds), Ablex, Nordwood NJ, 1993, p. 138-147.
- Klein G.A., "The fiction of optimization", *Bounded Rationality: The Adaptive Toolbox*, Gigerenzer G. and Selten R. (eds), MIT Press, MA, 2002, p. 103-121.
- Koch S., "Agile principles and open source software development", *Proceedings of the 5th Extreme programming and Agile processes in software engineering Conference*, Germany, 2004.
- Koontz H., O'Donnell C., Weihrich H., *Management*, McGraw-Hill. Tokyo, Japan, 1980.
- Kruchten P., "Agility with the RUP", *Cutter IT Journal*, Vol. 14, No. 12, 2001, p. 27-33.
- Levine L., Baskerville R., Link J., Pries-Heje J., Ramesh B., Slaughter S., "Discovery Colloquim: Quality software development at internet speed", *Technical Report CMU/SEI-2002-TR-020*, Software Engineering Institute, PA, USA, 2002.
- Lindvall M., Basili V., Boehm B., Costa P., Dangle K., Shull F., Tesoriero R., Williams L., Zelkowitz M., "Experical findings in Agile methods", *Proceedings of Extreme Programming and Agile methods – XP/Agile Universe Chicago*, 2002, p. 197-207.
- Lucas H., *Companion to Management Studies*, Heinemann Ltd. London, UK, 1982.
- Martin R., *Agile software development, Principles, patterns, and practices*, Prentice Hall, NJ, USA, 2003.
- Olson D., Stimmel C., *The manager pool, Patterns for radical leadership*, Addison-Wesley, NJ, USA, 2002.
- Orr K., "CMM versus Agile Development: Religious Wars and Software Development", *Agile Project Management*, Vol. 3, No. 7, 2002, p. 1-29.

- Pikkarainen M., Passoja U., "An approach for assessing suitability of Agile solutions: A case study", *Proceedings of Extreme Programming and Agile processes in software engineering (XP 2005)*, 2005, Sheffield, UK.
- Poole C. Huisman J., "Using extreme programming in a maintenance environment", *IEEE Software*, Vol. 18, No. 6, 2001, p. 42-49.
- Rasmusson J., "Introducing XP into Greenfield projects: Lessons learned", *IEEE Software*, Vol. 20, No. 3, 2003, p. 21-28.
- Reichheld F., "Lead for Loyalty", *Harvard Business Review*, Vol. 79, No. 7, 2001, p. 76-84.
- Reifer D., "How to get the most out of Extreme Programming/Agile methods", *Extreme Programming and Agile methods – XP/Agile Universe Chicago*, 2002, p. 185-196.
- Rising L., Janoff S., "The scrum software development process for small teams", *IEEE Software*, Vol. 17, No. 4, 2000, p. 26-32.
- Rogers E., *Diffusion of Innovations*, The Free Press: New York, USA, 1962.
- Sammon D., Adam F., "Towards a Model of ERP Software Selection - Widening the Debate", *The Enterprise Resource Planning Decade: Lessons Learned and Issues for the Future*, Adam F. and Sammon D. (eds.), Idea Publishing Group, Hershey, PS, 2004.
- Stake R., "Case Studies" Denzin, N., Lincoln Y.S. (eds) *The Handbook of Qualitative Research*, Sage Publications, CA, USA, 2000, p. 453-454.
- Stapleton J., *DSDM Business focused development*, Pearson Education, London, UK, 2003.
- Storey J., *Managerial Prerogative and the Question of Control*, Routledge&Kegan Paul, London, UK, 1983.
- Sutherland J., "Agile can scale: Inventing and Reinventing SCRUM in five companies", *Cutter IT Journal*, Vol. 14, No. 12, 2001, p. 5-11.
- Svensson H., Host M., "Introducing and Agile process in a software maintenance and evolution organization", *Proceedings of the 9th European Conference on Software Maintenance and Reengineering*, Manchester, UK, 2005, p. 256-264.
- Swanson E., "Telling an innovation story", *Diffusing software product and process innovations*, Ardis M., Marcolin B., (eds), Kluwer Academic Publishers, MA, USA, 2001, p. 1-4.
- Swanson E.B., Ramiller N.C., "Innovating mindfully with information technology", *MIS Quarterly*, Vol. 28, No. 4, 2004, p. 553-583.
- Toleman M., Ally M., Darroch F., "Aligning adoption theory with Agile system development methodologies", *Proceedings of the 8th Pacific Asia Conference on Information Systems*, Shanghai, 2004, p. 458-471.
- Turk D., France R., Rumpe B., "Assumptions Underlying Agile Software-Development Processes", *Journal of Database Management*, Vol. 16, No. 4, 2005, p. 62-87.
- Williams L., Kessler R., Cunningham W., Jeffries R., "Strengthening the case for pair-programming", *IEEE Software*, Vol. 17, No. 4, 2000, p. 19-25.
- Young G., "Implementing DSDM in eBA", *DSDM Business focused development*, Stapleton J. (ed), Pearson Education, London, UK, 2003, p. 97-103.